BLDRCODE LAB

TUTORIAL 1

INTRODUCTION TO PYTHON PROGRAMMING

Welcome to BLDRCODE Lab

Let's Get Started!

What is Python?

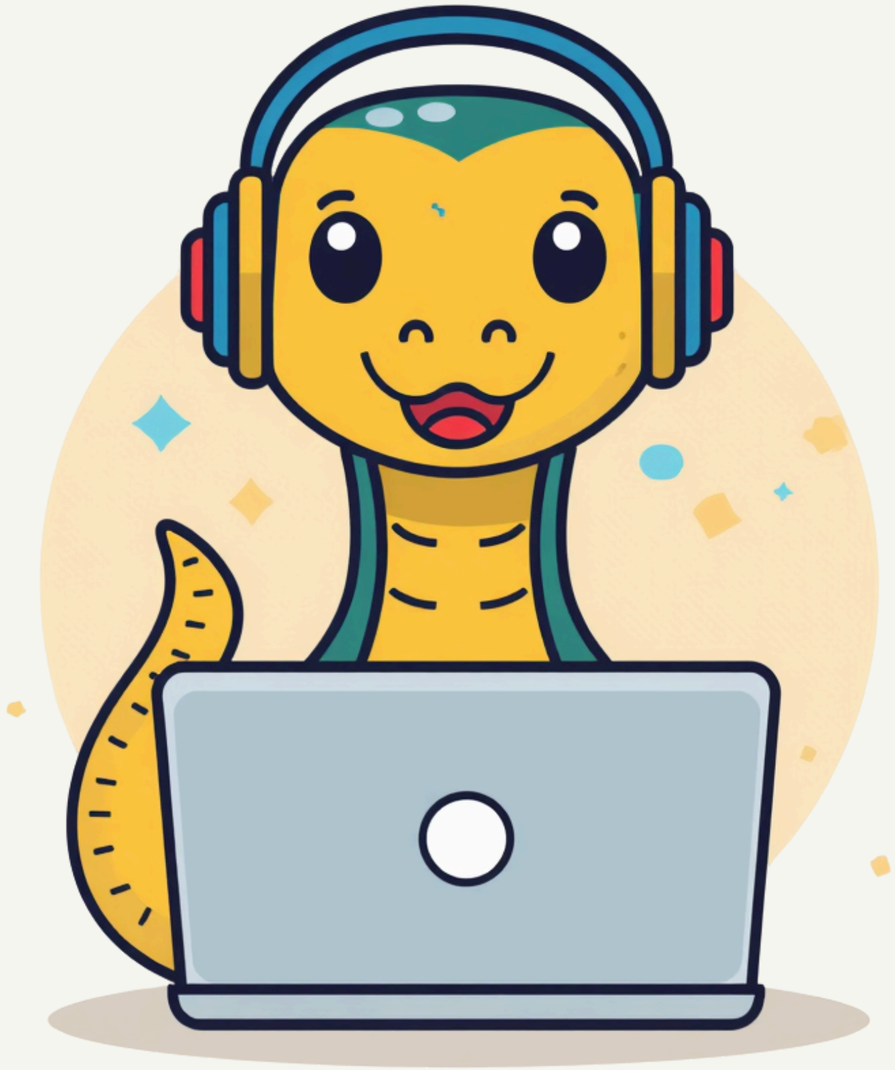
Definition & Uses

Python is a programming language that allows you to:

- Build software and applications
- Create games and interactive experiences
- Analyze data and visualize information
- Build AI and machine learning models
- Automate repetitive tasks
- Create websites and web applications

Fun fact: Python is used by Google, Netflix, Instagram, NASA, banks, and tech companies worldwide!

Key message: Python is one of the easiest and most powerful languages to learn.



Why Learn Python?

Simple Syntax

Python uses simple, clean syntax that reads almost like English. You can write powerful programs with fewer lines of code.

Used Worldwide

Python is used by millions of developers worldwide. It's perfect for beginners yet powerful enough for professionals at top companies.

Logical Thinking

Learning Python teaches you to think logically and break down complex problems into simple, manageable steps.

Installing Python

Three Easy Steps to Get Started

Step 01

Go to python.org and click the Download button. Make sure to select Python 3 (the latest version).

Step 02

Run the installer. Important: Check the box that says "Add Python to PATH" before clicking Install.

Step 03

Open Terminal (Mac) or Command Prompt (Windows) and type: python --version to confirm installation.

Running Python Code

Interactive Mode

Run python in your terminal to start the interpreter. Type print("Hello World") and press Enter to see instant results.

File Execution

Create a file called hello.py with your code inside. Run it by typing python hello.py in your terminal. Python executes line by line.

Basic Python Syntax

```
```

LET'S WRITE OUR FIRST PYTHON CODE! HERE'S A SIMPLE EXAMPLE:

Key concept: Python reads and executes code from top to bottom, one line at a time. Each print() statement displays text on the screen.

```
```

```
print("Welcome to BLDRCODE Lab")
print("We are learning Python")

Output:
Welcome to BLDRCODE Lab
We are learning Python
```

Variables in Python

Storing data in containers

What Are Variables?

Variables are like labeled containers that hold information. You give them a name, and Python remembers the value you store inside.

Text (String)

name = "John" stores text (called a string). Strings are always wrapped in quotes. Use them for names, words, or any text data.

Numbers & More

age = 15 stores a whole number (integer). country = "Ghana" stores another string. You can mix different types of variables in your programs!

Integers are whole numbers like 10, 25, or 100. Floats are decimal numbers like 1.5, 3.14, or 99.99.

INTEGERS & FLOATS

x = 10, y = 5

Python supports basic math operators: + (add), - (subtract), * (multiply), / (divide). Use them just like a calculator!

MATH OPERATORS

result = x + y → 15

Python can perform calculations instantly. Store numbers in variables, apply operators, and print results!

CALCULATE LIKE A PRO

Python = Calculator!

Mini Project – BMI Calculator

What is BMI?
BMI stands for Body Mass Index – a simple measure that uses your height and weight to estimate body fat.

The Formula:
BMI = weight / (height × height)
Example Calculation:
weight = 70 kg
height = 1.75 m
BMI = 70 / (1.75 × 1.75) = 22.86

Now let's build a Python program to calculate this automatically!

Interactive BMI Calculator

User interaction with input()

Now let's make our BMI calculator interactive! We'll use input() to get data from the user.

How it works:
- input() asks the user to type something
- float() converts text input to a decimal number
- The program waits for user responses before calculating

```
weight = float(input("Enter your weight in kg: "))
height = float(input("Enter your height in m: "))
bmi = weight / (height * height)
print("Your BMI is:", bmi)
```

Wrap Up

Next Steps

Thank You for Joining BLDRCODE Lab!

Ready to keep building with Python?

Questions?

CONNECT WITH US FOR MORE PYTHON COURSES

Our Mission

BUILDING THE NEXT GENERATION OF CREATORS AND PROBLEM SOLVERS

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